

THE CONDITIONAL IMPORTANCE OF PRIOR TIES: A GROUP-LEVEL ANALYSIS OF VENTURE CAPITAL SYNDICATION

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While there is growing scholarly interest in multiparty collaborations that involve three or more firms, most of this literature has focused on the governance, performance, and dissolution of such collaborations. Accordingly, less is known about which multiparty collaborations are likely to form or the extent to which the formation of multiparty collaborations may differ from the formation of dyadic collaborations. We address these gaps by exploring the formation of multiparty venture capital syndicates—a context that exhibits substantial multiparty collaboration but is not often examined through the multiparty lens. We argue and show that multiparty syndication is influenced by a number of *group-level* constructs. We highlight how group-level venture capital syndicate formation is not a simple deductive elaboration of dyadic venture capital syndication, but, instead, involves different constructs and logics. We also offer insight into how status dynamics may differ between dyads and groups.

Interfirm collaborations are an important mechanism by which firms obtain resources, manage uncertainty, and collaborate in the pursuit of goals (Gulati, 1995; Pfeffer & Salancik, 1978; Powell, Koput, & Smith-Doerr, 1996). While the corresponding interfirm literature has traditionally focused on the formation, governance, and performance of dyadic collaborations between individual pairs of firms, multiparty collaborations are also a common phenomenon; by some estimates, 30–50% of strategic alliances involve three or more partners (Garcia-Canal, Valdes-Llaneza, & Arino, 2003; Makino, Chan, Isobe, & Beamish, 2007). Such multiparty collaborations allow for a more diverse set of resources, experiences, and connections than dyadic collaborations, while also further distributing risk.

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Yet, multiparty collaborations exhibit greater complexity and different dynamics than dyadic collaborations in a number of ways (Das & Teng, 2002). As such collaborations depend upon a larger number of firms, each member can potentially hold up the entire group (Zeng & Chen, 2003), some partners may enter after or leave before others (Lavie, Lechner, & Singh, 2007), and the presence of three or more firms may foster the emergence of subgroups around faultlines (i.e., divisions within a team based on intra-group similarities and intergroup differences) (Heidl, Steensma, & Phelps, 2014). Thus, the differences between dyadic and multiparty collaborations go beyond simply the number of partners, rendering multiparty collaboration more than just a collection of dyads (Simmel, 1950). As Lavie et al. (2007) noted, multiparty collaboration is “multilateral interaction among partners and thus generates *unique* dynamics” (Lavie et al., 2007: 578; *italics added*).

Recognizing the uniqueness of multiparty collaborations, research has begun to offer clarity into the governance (Li, Eden, Hitt, Ireland, & Garrett, 2012), performance (Lavie et al., 2007), and dissolution (Heidl et al., 2014) of such collaborations, especially in the context of strategic alliances. Yet, the *formation* of

multifirm collaborations remains underexplored. Although some research has been done regarding the drivers of individual firms' decisions to participate in multiparty collaborations (e.g., Sakakibara, 2002; Li, 2013), we know very little about who partners with whom in multipartner settings.¹ It also remains unclear as to how the multiparty perspective may extend to interfirm collaborations other than strategic alliances.

Unlike collaborations that involve only two firms (Gulati & Gargiulo, 1999; Podolny, 1994; Sorenson & Stuart, 2001; Walker, Kogut, & Shan, 1997), analyzing the formation of multiparty collaborations as a series of individual dyads can be misleading, for several reasons. First, analyzing multiparty collaborations as a collection of independent dyads treats the involvement of other group members as given; this approach risks introducing endogeneity issues by failing to capture the likelihood that firms may simultaneously consider the entire set of other potential partners that would also be a part of the collaboration. Second, a dyadic perspective leaves out group-level constructs that might influence the formation of multiparty collaborations. For instance, what might be the role of group-level attributes such as a potential group's density of prior ties, number of potential members, network-based faultlines, or status-based disparities? Since many of these constructs only arise at the group level, their influence on the formation of multiparty collaborations is not a simple deductive extension of dyad-level theory and findings. In short, treating the formation of multiparty collaborations as a set of independent dyads fails to capture the full complexity and challenges of such collaborations (Albers, Schweiger, & Gibb, 2015).

In this paper, we seek to address these gaps by studying the formation of multiparty investment syndicates among venture capital (VC)² firms. VC

investment is an important economic and social phenomenon, funding ventures that have an innovative, high-growth, and disruptive impact in many industries (Gompers & Lerner, 1999; Gupta & Sapienza, 1992). For ventures, VC funding adds value by increasing the adoption of professional practices, providing a signal of legitimacy, and building a network of connections (Hallen, 2008; Hellmann & Puri, 2002). Within our research, one of the most relevant aspects of this phenomenon is that venture investing is often a "team sport"³—that is, in a majority of the cases, investment in new ventures is done by a syndicate of VC firms rather than by VCs investing solo.

VCs participate in syndicated deals for a variety of reasons, including to better evaluate ventures by having multiple judgments on a deal, to better aid ventures by assembling broad and complementary networks and areas of expertise, to share risk, and to develop reciprocal relationships that provide future access to high-quality deals—especially outside of their focal industry or geography (Brander, Amit, & Antweiler, 2002; Bygrave, 1987; Hochberg, Lindsey, & Westerfield, 2015; Hochberg, Ljungqvist, & Lu, 2007; Jääskeläinen, 2012; Lerner, 1994; Sorenson & Stuart, 2001, 2008). Consistent with broader studies of interfirm collaborations (Gulati & Gargiulo, 1999; Uzzi, 1997), syndication exhibits substantial social embeddedness as VCs syndicate primarily with past syndication partners so as to better identify trustworthy and high-quality partners, as well as to create a barrier to entry (Hochberg, Ljungqvist, & Lu, 2010; Kogut, Urso, & Walker, 2007; Sorenson & Stuart, 2008). Recent research has also begun to explore the drivers of exceptions to this tendency, finding that prior ties are less important for syndication when VCs are more familiar with one another through prior competition (Trapido, 2007) and when agency risks are lower (Meuleman, Lockett, Manigart, & Wright, 2010). Yet, prior VC syndication studies have taken a dyadic perspective, both theoretically and empirically, in studying partner selection, leaving unclear how group-level constructs might also influence which VC syndicates are more likely to form.

In an attempt to begin moving VC syndication research from the dyad to the group level and to better understand the formation of multiparty alliances more broadly, this paper presents a group-level analysis of syndication among VC firms. Focusing on the formation of syndicates comprising three or

¹ Two relatively recent studies have focused squarely on partnership *formation* in multiparty alliances, but do not analyze such partnerships from a group-level perspective. In their study of partnerships in shipping alliances, Mitsuhashi and Greve (2009) focused on the decisions of individual firms to join an alliance, rather than on the emergence of the entire partnership as a group-level phenomenon. And, in their analysis of the formation of international governmental organizations involving multiple states, Ingram and Torfason (2010) computed group-level attributes by averaging across dyads within the group, rather than by looking at the relevance of group-level attributes which would not have any meaning at a dyadic level.

² Consistent with common parlance in the industry, we use the term "VC" to refer to venture capital and venture capitalists as well as venture capital firms.

³ The term "team sport" comes from one of the VCs whom we interviewed during the early phase of this study.

more VCs at the time of first-round investments in the venture, we address the following question: *How do prior co-investment ties among a particular group of VC firms shape whether or not they form an actual multiparty syndicate at the time of first-round investments in a new venture?*

Although we test our arguments and hypotheses by analyzing data on all first-round multiparty VC syndicates formed during the period 1985–2008, in the first phase of our research, we conducted field interviews with both VCs and entrepreneurs (12 each) in order to better understand the particular concerns and challenges surrounding the formation of syndicates with three or more VC firms. We also draw on fieldwork on VC syndication previously conducted by one of the co-authors as part of a study on entrepreneur fundraising (Hallen & Eisenhardt, 2012). Like all high-stakes collaborations (Gulati, Wohlgezogen, & Zhelyazkov, 2012), our fieldwork reveals that syndicating VCs have both *cooperation* concerns about the motivations guiding other VCs in a potential syndicate as well as *coordination* concerns around the complexity of structuring joint and interdependent action (including the initial decision to jointly syndicate). By explicating the specific nature of cooperation and coordination issues in VC syndication, our fieldwork helps offer insights into when such concerns may be more or less pronounced. Additionally, our fieldwork also reinforces the group-level nature of the formation of multi-VC syndicates, as we observe that each VC is generally concerned about every other potential syndicate member's involvement and future behavior.

We build on insights from our fieldwork and extant theory to deductively develop hypotheses about how group-level characteristics and industry attributes influence the formation of syndicates involving three or more VCs. We argue that, while offering many benefits, group-level attributes and contextual factors such as syndicate size or industry popularity may also lead to cooperation and coordination challenges. Drawing on longstanding arguments that repeated tie density provides various forms of social capital such as trust, reciprocity, and social identification (Coleman, 1988, 1990; Dyer & Singh, 1998; Phelps, 2010) and that social capital plays a role in VC syndication (Hallen, 2008; Sorenson & Stuart, 2001), we start with the baseline premise that a group of VCs is more likely to syndicate a given deal together when more of the VCs have previously worked together such that there is a higher level of tie density within the potential

syndicate. Building on this premise, we theorize about how various group-level and industry-level attributes might make the cooperation and coordination challenges more or less pressing, amplifying, or dampening the importance of tie density. We theorize that the positive impact of tie density on the likelihood of syndicate formation is greater when a potential syndicate is larger, when it suffers from higher network-based fault line risk, and when there is greater disparity in the status of the VCs. We also examine how contextual factors such as industry popularity may have a similar effect.

To better highlight group formation (vs. group evolution), we focus only on first-round investments whereby all investors are new to a deal and simultaneously co-investing. We quantitatively test our arguments using a novel empirical approach contrasting 1,336 real and 6,680 matching hypothetical syndicates of three or more VCs at the time of first-round investments in U.S.-based ventures during the period 1985–2008. Our analyses reveal both general support for our arguments and unexpected nuances.

This paper makes several important contributions. First, it offers a more complete perspective than the prior dyadic approach on how VC firms form syndicates by highlighting the importance of group-level dynamics in multiparty VC syndicate formation. Second, it contributes to the broader network formation literature by explicating how group-level formation is not a simple deductive elaboration of dynamics at the dyadic level, as we find that multiparty collaboration is strongly influenced by several group-level constructs that have no clear analogy in dyadic collaborations (e.g., syndicate size, network faultlines, and status subgroups). Third, we show how status dynamics may differ between dyads and groups, as VC syndicates are more likely to form among VCs of different (rather than similar) status—especially in the case of syndicates with high network density.

VENTURE CAPITAL SYNDICATION

Data from VentureXpert (Thomson Reuters) for all VC investments during 1985–2008 indicate that multiparty syndicates (i.e., those involving three or more VCs) account for 60% of all syndicated investments. For first-round investments (the focus of this study), the proportion of multiparty syndicates is only slightly less (i.e., 44%). Clearly, multiparty syndication is a highly common occurrence in the case of VC investments. Building on extant research

as well as original field interviews, we discuss, below, why syndication occurs, how syndicates are formed, and the various cooperation and coordination challenges therein.

VC syndication can potentially yield six different types of benefits. First, syndication often helps in making better selection decisions and, post investment, better outcomes for the ventures; this happens because it enables the pooling of diverse and at least partially complementary judgments and capabilities of multiple VCs (Brander et al., 2002; Tian, 2011). Second, syndication can enable a VC firm to reduce risk by allocating available capital across a larger number of ventures (Bygrave, 1988). VC investments are a high-risk activity, and, since the fund size is always finite, a VC firm can use syndication to invest in a larger number of ventures, thereby lowering the overall risk. As one VC put it, syndication allows his firm to “bet on more horses.” To quote another VC:

“Could we have done the whole round ourselves? We could have, but [getting other VCs in] is advantageous in some ways as it gets another seat around the table with the assumption that you’re going to need more capital in the future and share some of the cost burden [of follow-on investments].”

Third, syndication can provide VC firms with access to a larger pool of future opportunities (“deal flow,” as one VC put it) that syndicate partners may bring to them (Manigart et al., 2006). Fourth, syndication can enable VC firms to invest in nonlocal ventures by partnering with other VC firms for which this would be a local investment (Sorenson & Stuart, 2001). Fifth, syndication can enable a VC firm to “window-dress” the portfolio by showcasing the status of co-investing partners (Lerner, 1994). Sixth, syndication can help erect entry barriers against new VCs thereby yielding better investment terms for the syndicate partners (Hochberg et al., 2010).

Prior literature has sometimes portrayed the formation of a VC syndicate as a cascading process whereby the entrepreneur comes to an agreement with a lead VC who then invites other VCs to join as co-investors. Were this to be the case, then multiparty VC syndicate formation might be approximated as the formation of otherwise independent dyads with a common lead VC. Our interviews reveal, however, that this perspective is often inaccurate. Many entrepreneurs initiate simultaneous discussions with several VCs in order to get the best deal and find the best fit. Some VCs may then encourage the entrepreneur to talk to other VCs in their

network. As potential investors discover who the other co-investors may be, they may remain interested, may become more interested, or may drop out. This is also true for non-lead VCs, in those cases in which there is a lead VC setting up the syndicate. Syndication formation also requires agreement around valuation and deal terms (since all VCs invest at the same price), who gets to invest how much (and thus who gets how much equity), and board seats and other roles. Sometimes, interested VCs may come together to form a syndicate (thus reducing competition for the deal), while, at other times, competing syndicates may emerge (allowing VCs greater discretion over syndicate partners). Even while discussing the deal collectively, VCs may still conduct substantial due diligence independently. Each must also reach agreement within their own firm partnership about whether to invest. Even preliminary agreements regarding “lead” roles can fall apart. Potential syndicates can even fall apart after a term sheet has been offered but the money has not yet been delivered to the entrepreneur (the round closing).

This mutual agreement and dynamic nature of multiparty syndicate formation is illustrated in the following quotes from our interviews:

FOUNDER/CEO, VENTURE A: “The problem with [KLM, a top-tier VC firm that eventually chose not to invest] was that they were always very interested in us but they weren’t interested in any of the other VCs that we were talking to.”

FOUNDER/CEO, VENTURE B: “[XYZ] came in and said, ‘We’re taking \$1.5 M of the \$2.5 M left.’ The partner from [ABC, the other VC firm] said, ‘Bullshit. I assumed we were going to split the last \$2.5 M, \$1.25 M each.’ The guys at [XYZ] then made up some bullshit story about their investment community saying they have to put in at least \$1.5 M.”

FOUNDER/CEO, VENTURE C: “There was a larger VC firm who wanted to invest but couldn’t get the [decision] down. Once I told them that the term sheet [from other investors] was coming, the partner that I was working with said that he [needed] another six weeks or so. So I said I’m going to move on.”

FOUNDER/CEO, VENTURE D: “A term sheet isn’t a guarantee to the money. More deals fall apart [trying to agree on final investment terms] than any other way.”

As a whole, and to paraphrase one of the VCs whom we interviewed, a syndicate *crystallizes* only when there is clarity and agreement about who will invest, how much each party will invest, the allocation of ownership stakes, and various formal rights

(to information, to a board seat, to an observer status, and so forth). Until then, willingness to invest is merely an expression of interest, not a commitment.

In terms of the factors that may shape the composition of a syndicate, our interviews reveal that potential partners are highly sensitive to mutual complementarity, the motivations guiding each potential co-investor, and the likelihood that the various parties will be able to work together productively after the investment. To quote from our interviews:

VC: "Whom to syndicate with? First, someone we had a relationship with, whom we knew and respected. Someone that brought something incremental to the table. You look at the person involved. You look at where they are in their fund. You look at how successful they've been in the past. . . . It comes down to chemistry. Do you want to work with the people? And obviously they have to want to do it too."

These observations are consistent with Gulati et al.'s (2012) broader analysis of collaboration among organizational actors. As they noted, forward-looking concerns about future cooperation and coordination challenges can have a direct bearing on ex ante decisions regarding with whom to collaborate. *Cooperation challenges* pertain to misalignment in incentives, whereas *coordination challenges* pertain to the mechanics of working together to achieve the hoped-for benefits. Notwithstanding their distinct roots, cooperation and coordination often complement and reinforce each other. Greater alignment in incentives may lead actors to make greater investments in coordination mechanisms, and vice versa (Gulati et al., 2012).

In the context of VC syndication, cooperation challenges can emerge on account of several factors. First, VCs may differ in their risk propensities such that some prefer the venture to pursue a higher-risk strategy with the goal of achieving a "home run," whereas others may prefer the pursuit of a lower-risk strategy that aims at "singles." Second, different VCs may have differing proclivities along a number of key dimensions, such as the relative importance of the team ("the jockey") versus the opportunity ("the horse"), the advisability of patenting versus not patenting, and how perfect versus imperfect the product/service should be before it is launched in the market. Finally, the potential or actual syndicate may suffer from a high risk of faultlines that get in the way of arriving at optimal decisions in a timely manner (Heidl et al., 2014).

We focus now on the coordination challenges that potential syndicates may face. First, there may be

occasions that require rapid decision-making on the part of potential investors about whether or not to invest. VCs without a history of prior collaboration may be slower to come together and become comfortable with both the deal and one another, as compared with those who know each other from prior syndication. Second, VCs typically have extremely busy calendars, and, as is well known, consider time to be their most scarce resource (Freeman, 1999). The larger the syndicate size, the harder it may be to reach agreement on investment terms and to organize in-person meetings for rich discussions. Third, a similar coordination challenge may be faced if, in terms of their geographic locations, the VCs are distant from one another. Finally, different VCs may differ in their governance styles. Prior research tells us that some VCs tend to be "close trackers" whereas others prefer a more "laissez-faire" approach (Macmillan, Kulow, & Khoylian, 1988). Thus, in any given situation, some VCs in a syndicate may believe that an early meeting of the investors is warranted whereas others may not sense such urgency.

In sum, while VC syndicates can potentially yield many benefits, they can also be prone to a number of cooperation and coordination challenges. Our premise is that decisions regarding which VC firms come together in a syndicate will be a function of the size and salience of the potential benefits and challenges associated with the composition of the prospective syndicate.

THEORY AND HYPOTHESES

Our overarching argument is that various group-level attributes and contextual factors may provide benefits and yet may also introduce coordination and cooperation challenges in the formation of multiparty VC syndicates. Accordingly, we expect the impact of such factors to depend on their interplay with the *tie density* amongst the potential members of a VC syndicate. We define "tie density" as the number of repeated ties within a prospective group divided by the number of all possible dyadic ties among the group members (Wasserman & Faust, 1994). Tie density aggregates the extent of tie repetition among all dyads, and can range from complete novelty at one extreme (i.e., no tie repetition in any dyad) to complete closure at the other (i.e., tie repetition in every dyad). Such density thus builds on the dyadic construct of tie repetition, a binary construct capturing whether or not two firms (VCs) have previously cooperated with each other; however, tie density extends dyadic tie repetition by recognizing

that the prospective group may include a combination of repeated as well as novel ties.

Accounting for tie density is important as it provides various forms of social capital (trust, reciprocity, and social identification) that may enhance interorganizational cooperation and coordination (Coleman, 1988, 1990; Dyer & Singh, 1998; Phelps, 2010). Indeed, prior research from a dyadic perspective on VC syndication has shown a strong tendency toward syndication where tie repetition is present (Sorenson & Stuart, 2008). Building on this literature, our basic premise is that multiparty VC syndication will also be substantially influenced by tie repetition. That is, all other things being equal, tie density among a group of potential co-investors will have a positive impact on the likelihood of actually forming a new multiparty syndicate.

Theoretically, higher tie density may offer positive benefits through three generally complementary mechanisms. First, at higher levels of tie density, there would be a greater number of repeated ties within the syndicate; such ties elevate mutual trust and understanding among individual pairs of group members who have collaborated before (Gulati, 1998; Kogut et al., 2007; Rowley, Behrens, & Krackhardt, 2000). Second, in groups with intermediate levels of tie density, even group members who may not have direct experience with one another are more likely to be connected to each other through indirect ties; such indirect ties provide information and legitimacy, thereby reducing any concerns that indirectly connected potential group members may have about one another (Gulati, 1998; Sorenson & Stuart, 2001; Walker et al., 1997). Third, higher tie density is associated with greater network closure among group members, such that group members may not only have prior direct ties but also have many prior common partners who are also members of the group. Such closure can be especially effective in fostering trust and high collaborative effort, as group members may be hesitant to endanger not only their direct relationships but also relationships with common partners (Ahuja, 2000; Burt, 2005). Together, these arguments suggest that greater tie density in syndicates will reduce cooperation concerns by lowering the likelihood of opportunism, and may aid coordination by making involved VCs more willing to rely on one another's due diligence and accelerating the speed with which they may decide whether or not to make an investment.

Notwithstanding the possible benefits from repeated ties, we argue that VCs face a mixed-motive

situation. While relying on prior ties can make cooperation and coordination easier, it also inhibits an extension of a VC's network. The latter motive is extremely important in situations where access to and exploration of new opportunities is at least as critical as exploitation of current ones.⁴ As a VC we interviewed put it:

"The ecosystem around us is constantly in flux—with a never-ending stream of new technologies, new market opportunities, and new business models. We can't be experts at everything. Thus, much as we like to work with VCs whom we know well, we also look for opportunities to expand our network and knowledge base. Co-investing with new partners helps us do this. It also helps us stay in the middle of a larger deal flow in the future."

Another factor that leads VCs to join syndicates with at least some novel ties is that it may be hard for VCs to find a set of syndication partners with whom they have prior ties and who are also interested in a deal.⁵ For instance, it may be the case that other VCs with whom a VC has prior ties may already have invested in competing ventures, thereby creating conflicts of interest among otherwise desirable syndicate partners (Pahnke, McDonald, Wang, & Hallen, 2015).

Given the contrasting effects of prior ties, we theorize that the salience of prior ties in the formation of multifirm syndicates will vary across contexts. More specifically, higher tie density and the social capital benefits it provides will be deemed more important when contextual conditions make the emergence of effective collaboration among the co-investors more challenging and/or more urgent. In this study, we focus on four contextual conditions: (1) syndicate size, (2) risk of network faultlines, (3) status disparities within the syndicate, and (4) industry popularity. The first three conditions pertain to the internal context of a potential syndicate whereas the last condition pertains to the external context. Our choice of these four conditions derives from three premises. First, these are among the most important factors that VCs look at when deciding whether or

⁴ Contrast the case of co-investment partnerships by VCs with that of supply chain partnerships by auto firms. Both are multiparty alliances. However, the need for ongoing exploration is much greater in the former context than in the latter.

⁵ We thank an anonymous reviewer for encouraging us to also consider the numerical challenges of forming dense syndicates.

not to form a syndicate.⁶ Second, the first three are group-level factors, thereby helping us examine the formation of VC syndicates from a uniquely group-level (vs. dyadic) lens. Third, all four have a direct bearing on the ease or urgency of effective cooperation and coordination among the co-investors.

We now elaborate on how we expect each of these contextual conditions to alter the relative importance of prior tie density in the formation of multiparty VC syndicates. Our unifying argument is that prior tie density may often reduce the cooperation and coordination challenges associated with each of these factors, thus altering the balance between each factor's benefits and drawbacks.

Syndicate Size

Multiparty VC syndication deals exhibit considerable variance in syndicate size. According to VentureXpert data, during the period 1985–2008, for all multiparty VC syndication deals in the United States, syndicate size equaled three in 40% of the cases, four in 23% of the cases, and greater than or equal to five in 37% of the cases. For first-round multiparty syndicates (the focus of our study), the variance was somewhat less, but still fairly high; for such deals, syndicate size equaled three in 59% of the cases, four in 23% of the cases, and greater than or equal to five in 18% of the cases.

Our field interviews as well as blog posts⁷ by VCs reveal that syndicate size in multiparty rounds is determined by a combination of several factors. Venture CEOs—typically, the dominant owners and decision-makers prior to first-round VC investments—have several reasons to prefer a larger syndicate. A larger syndicate would result in smaller ownership stakes by individual VCs; this would give the venture CEO greater relative power vis-à-vis any single VC (Garg,

2013). In our field interviews, this emerged as a key consideration for some entrepreneurs, especially those with a strong orientation toward retaining control over the venture (Wasserman, 2012). By relying on multiple coinvestors, a larger syndicate also makes it easier for the entrepreneur to raise the initial sum of targeted investment. More importantly, a larger syndicate significantly reduces the risk of not being able to secure needed investments in the second or later rounds. A larger syndicate also increases the venture's ability to tap into a larger pool of non-overlapping expertise and network linkages.

From the VCs' perspective, a larger syndicate presents a mixed-motive situation. While a larger syndicate may help them extend their network, it also dilutes their decision-influencing power over the venture and may often reduce the equity (i.e., the percentage of a venture) that any one VC may purchase (Suster, 2011). The biggest constraint to a larger syndicate derives, however, from the rapid increase in cooperation and coordination risks that emanate from larger syndicate size. As syndicate size increases, the number of dyadic relationships increases geometrically. Conflicting viewpoints and/or misalignment of interests in any of these dyads can become an impediment to effective cooperation. In a widely followed blog post (Feld, 2009), a seasoned VC highlighted the cooperation risks that may accompany larger syndicate size thus:

“As the syndicate size grows, the chance of interests between parties getting out of alignment increases. . . . When everything is going well, this isn't an issue, but the minute the business goes sideways (or worse) strange things start to happen. As the situation degenerates, the knives (or flamethrowers) come out. I've been involved in situations that resulted in the destruction of companies that deserved to live another day because the investors around the table (which included me) couldn't get their collective shit together.”

In other words, as the number of syndicate members increases, the chance of at least one of the co-investing VCs subsequently causing problems also increases.

A larger syndicate can also increase coordination risks. Prospective members of a larger syndicate may find it harder to agree in a timely manner on not only whether they will invest in the venture but also the terms that they should offer. If the offer is accepted and the investment is made, a larger syndicate would generally result in a larger board of directors and more board observers. A larger-sized board may find

⁶ This is borne out in our field interviews. When forming a syndicate, the factors that VCs deem to be most salient include the size of the syndicate, the identity of other potential co-investors, and the external context within which the investment decision is to be made. Industry distance and geographic distance are two other factors that are very salient for investors. We control for these factors but do not include them in our theoretical analysis, for the reason that past studies have already analyzed the effects of these factors (Sorenson & Stuart, 2001, 2008).

⁷ Blog posts have become a prevalent mechanism by which VCs seek to establish their reputation as thought leaders and domain experts amongst entrepreneurs. A common topic of these blogs is educational posts explaining VC decision-making and dynamics.

it more difficult to arrange in-person board meetings; this is important because reliance on telephonic meetings will often reduce the richness of discussions, thereby negatively affecting the quality of decisions. A larger-sized board also increases the risk of conflicts arising from incompatible personalities. As one VC noted, “All of us have high egos. You really have to watch out for personality and ego conflicts.”

Our arguments regarding the collaboration risks that accompany larger VC syndicates are consistent with the broader set of research findings about the cooperation and coordination challenges faced by larger teams or larger-sized multiparty interfirm alliances (Amason & Sapienza, 1997; Colquitt, Noe, & Jackson, 2002; Yin & Shanley, 2008). It is these challenges that led Zeng and Chen (2003) to argue that larger interfirm alliances are likely to be less stable than smaller ones.

We theorize that prior tie density can play a compensating role in mitigating the collaboration risks that accompany larger syndicate size. As argued earlier, prior ties, common third parties, and network closure will increase mutual understanding and trust among the syndicate partners. They also ensure the availability of preexisting communication links and behavioral norms. Given the greater collaboration risks faced by larger syndicates, the marginal impact of prior tie density can be expected to be stronger in the case of larger syndicates. We thus expect greater tie density to reduce the drawbacks of larger syndicates (e.g., more challenging coordination and cooperation) while still preserving their benefits (e.g., greater resources, networks, etc.).⁸ More formally stated:

Hypothesis 1. Prior tie density will have a stronger positive impact on the formation of multiparty VC syndicates when the size of the potential syndicate is larger.

Risk of Network Faultlines

Research on teams of individuals reflects a growing focus on the risk of demographic faultlines—that

is, divisions within a team based on intragroup similarities and intergroup differences in demographic and other factors such as age, gender, ethnicity, and education (Earley & Mosakowski, 2000; Lau & Murnighan, 2005; Li & Hambrick, 2005). In broad terms, the core conclusions of this research are that faultlines have a negative impact on team effectiveness; however, the negative impact is moderated by factors such as team identification and crosscutting actors who serve as bridges across the faultlines (DiBenigno & Kellogg, 2014; Mas, Flache, Takacs, & Jehn, 2013).

Heidl et al. (2014) is perhaps the only study that has extended the notion of possible faultlines from teams of individuals to formal collaborations among organizations. Network faultlines refer to a situation in which a division appears in group collaboration due to the presence of internal “subgroups” with strong network connections (i.e., prior relationships) to one another but only weak or no connections to those outside of the subgroup. Under such conditions, partners connected by strong ties may form an in-group while relegating other partners to out-groups. Heidl et al. (2014) argued that the presence of network faultlines would prevent alliance partners from investing the time and effort necessary to establish broad-based trust and effective decision-making routines *among all partners*. Focusing on multipartner alliances in the global telecommunications industry, the authors found support for their prediction that dispersions in tie strength among alliance partners would result in greater hazards of unplanned dissolution.

We build on Heidl et al.’s (2014) study and argue that multiparty VC syndicates can also suffer from network fault line risks. As one of the VCs in our field interviews commented:

“If we have to go to bed with other VCs, it’s just super critical that we respect them, trust them, and be able to work with them. Take ABC [another VC]. We’ve worked together on many deals over the last ten years. We know their quirks, what they like and what’ll tick them off. It’s easy to work with them. There are times, however, when the entrepreneur may insist on including one or more investors with whom we have had no relationship in the past. I’m always wary of such situations. Will they come out of the left field? Discussions are important. But, at the end of the day, we need a consensus not a gridlock.”

As we have argued earlier, a VC syndicate is not a bunch of loosely connected organizational actors. VCs are actively involved in almost all key venture decisions—especially in early-stage

⁸ It is important to note that, in order for VC firms to be able to form larger syndicates with greater density, on purely numerical grounds, each VC firm must have a large enough number of prior direct ties. Our analysis shows that the average number of direct ties for VCs in each focal year during the period 1985–2008 is in the range of 19–42, well above the focal syndicate sizes of 3–11 VCs (the majority of the syndicates in our sample involve 3–5 VCs).

ventures (Sapienza, 1992). As equity owners, all VCs have a right to participate in these decisions and they depend on one another to come to better decisions in a timely manner. In the absence of compensating mechanisms, network faultlines create the risk of weak cooperation and coordination. For the venture, this can mean lost time as well as suboptimal decisions.

We theorize that high prior tie density can serve as a compensating mechanism, thereby mitigating the risks emanating from network faultlines.⁹ A web of prior ties among VC firms increases the likelihood that there exist at least some bridges and that they have already built at least some degree of mutual understanding, trust, and shared decision-making routines. The greater the risk of network faultlines, the greater would be the importance of these compensating benefits emanating from a web of prior ties. We thus expect VCs to be less deterred by network faultlines when tie density within the potential syndicate is greater. More formally:

Hypothesis 2. Prior tie density will have a stronger positive impact on the formation of multiparty VC syndicates when the prospective syndicate suffers from a higher risk of network faultlines.

Status Disparities Within the Syndicate

As Washington and Zajac (2005: 284) noted, status is “a socially constructed, intersubjectively agreed-upon and accepted ordering or ranking of individuals, groups, organizations, or activities in a social system.” Status homophily has long been recognized as an important driver of social interactions. Given choice, people typically interact with others who are similar to them along a number of characteristics, including status (Blau, 1977; Homans, 1950). As Homans (1950: 184) observed, “The more nearly equal in social rank a number of men are, the more frequently they will interact with

one another.” The importance of status homophily derives from several mechanisms. Individuals of similar status are more likely to share similar meanings, attitudes, and beliefs; they are more likely to be empathetic with one another; and, when they interact, they are less likely to risk loss in status (Blau, 1977; McPherson, Smith-Lovin, & Cook, 2001).

Podolny (1994) broadened the importance of status homophily to economic exchanges between organizations. He argued that organizations also tend to partner primarily with others of similar status due to assortative matching. This comes about as high-status organizations restrict their matches to other high-status organizations due to underlying quality concerns, leaving lower-status organizations little choice but to partner with other lower-status organizations. Arguing that status serves as an important signal of quality, he found support for predictions that the propensity of organizations (such as investment banks) to partner with others of similar status would be stronger when they face greater market uncertainty.

Our field interviews suggest that organizational status is extremely important for VCs also. Given very high uncertainty, they are keen to take into account all relevant signals of venture quality, including the status of other VCs in the potential syndicate. While literature on syndicates with only two VCs has yielded some evidence of status homophily (Lerner, 1994),¹⁰ our fieldwork suggests that, in the case of multiparty syndicates, a number of countervailing factors may tilt the balance toward status disparities rather than homophily.

Consider first the barriers faced by prospective syndicates comprising multiple high-status VCs. Since high-status VCs typically manage larger funds, they prefer to invest a larger sum of capital and take a larger share of the equity in the venture. Such an outcome would be infeasible if a multiparty syndicate were to include several high-status VCs, especially for first-round investments. In such investments, the combined equity stake of all external investors tends to be no more than 25–30% (so as to safeguard the incentives for the founders and employees, given the almost-certain dilution in their stakes during later investment rounds). Second, because high-status VCs

⁹ Note that dispersion in tie strengths and tie density are distinct constructs. Consider two syndicates, both with three VCs. In Syndicate 1, A and B have co-invested with each other 10 times; however, neither has co-invested with C. In Syndicate 2, D and E too have co-invested with each other 10 times; however, each has also co-invested with F one time. In both syndicates, dispersion in tie strengths is almost equally high. However, in Syndicate 1, tie density = 1/3 (i.e., quite low), whereas, in Syndicate 2, tie density = 1 (i.e., very high).

¹⁰ While Lerner (1994) did not directly study status as captured in the organizational literature, he found evidence of experienced-based homophily in VC syndicates, which may be expected to be highly correlated with status homophily.

have larger portfolios of ventures in which they already hold equity, there is greater likelihood of their having already invested in a competing venture; thus, conflict of interest would prevent several high-status VCs from joining the same multiparty syndicate (Pahnke et al., 2015). Third, high-status VCs value being viewed as having played the primary role in a venture's development; this may be difficult if the syndicate were to include other VCs of equally high status. To quote from the observations of a prominent VC (Jones, 2008):

"VC deal syndication is a pretty funny dance, with the entrepreneur trying to get VCs to converge, and venture capitalists trying to feel out each other and the company to see how/where/when an investment might come together. . . . Venture capital firms are a bit like chickens, always concerned with their pecking order and social status . . . [they] worry about . . . who gets mentioned first in the press release."

Taken as a whole, these dynamics suggest several challenges to multiple high-status VCs co-investing in the same syndicate.

In contrast, multiparty syndicates comprised of VCs of disparate status may reduce many of these challenges. Status disparities make it easier for high-status firms to obtain a sufficiently high percentage of equity in the first round to justify their time and attention and enable them to maintain their high ownership stake through subsequent rounds. In fact, higher-status VCs can expect to further increase their ownership stakes in later rounds, since lower-status VCs tend to have smaller funds and subsequent rounds tend to require substantially greater capital investments than the first round. While these dynamics do suggest that lower-status VCs may receive less equity and have less influence over key decisions, they may be open to accepting such disparities, as they stand to gain from the status accrual that may result from partnering with higher-status VCs (Cowen, 2012). While the underlying drivers may differ somewhat, prior studies indicate that status disparities are also the norm in many other inter-firm alliance contexts, such as Formula One racing (Castellucci & Ertug, 2010) and the global chemical industry (Ahuja, Polidoro, & Mitchell, 2009). As Ahuja et al. (2009: 945) argued, for higher-status firms, an important benefit from partnering with lower-status firms is that the latter "will often offer favorable terms of trade to entice central allies into an exchange relation."

Notwithstanding the above-noted benefits that may attract high- and low-status VCs to one another

in multiparty syndicates, they must also contend with the cooperation and coordination challenges that may accompany status disparities. These challenges can manifest in two quite different ways: *status heterogeneity* and *status-based subgroups*. The first, status heterogeneity, refers to differences in status among the VCs in a prospective syndicate. Given a larger portfolio of ventures, high-status VCs are less dependent on the success of any particular venture; thus, they are likely to be less risk averse than lower-status VCs in any single deal. Our interviews also suggest that higher-status VCs tend to have a higher proportion of individual partners with direct entrepreneurial experience; in contrast, lower-status VCs tend to rely more heavily on individual partners with purely financial backgrounds. In sum, status heterogeneity increases the risk that the VCs in the syndicate will bring different preferences and cognitive lenses when evaluating the venture's performance and deciding what to do next.

The second—status-based subgroups—refers to the potential emergence of subgroups whereby members of one subgroup are similar in status to one another but different from other subgroups. Analogous to network-based faultlines, status-based subgroups may also give rise to faultlines. However, unlike network-based faultlines, these would be rooted in status similarities rather than strength of network ties. As with network-based faultlines, status-based subgroups are likely to dampen interaction and create lack of trust across the subgroups. Status-based subgroups are also conceptually and empirically distinct from the construct of status heterogeneity. For example, a syndicate in which all VCs have very different status levels would be characterized as having high status heterogeneity but no status-based subgroups.¹¹

We anticipate that higher tie density is likely to reduce the cooperation and coordination concerns associated with both forms of status disparities in multiparty syndicates. Prospective syndicates with higher tie density will have a larger proportion of both direct as well as indirect ties. These preexisting

¹¹ Consistent with our arguments that status heterogeneity and status-based subgroups are distinct constructs, our empirical analysis indicates that the two are not positively correlated. In fact, the two constructs exhibit a slight negative correlation ($r = -.13$). We interpret the (slight) negative correlation as suggesting that, as the presence of status subgroups increases, the level of status heterogeneity decreases (since VCs within a subgroup are similar to one another in status).

ties can help bridge gaps in mutual understanding and trust, alleviating cooperation and coordination challenges in a multiparty syndicate. In other words, greater tie density may reduce the drawbacks of status disparities in VC syndicates (e.g., differences in preferences and cognitive lenses, and the risk of poor communication and mistrust) while still preserving the benefits (e.g., allowing higher-status firms to acquire greater equity and lower-status firms to improve their status). It follows that the stronger the possible collaboration hazards arising from status disparities among the VCs in a prospective syndicate, the greater would be the value of preexisting ties. In more formal terms:

Hypothesis 3a. Prior tie density will have a stronger positive impact on the formation of multiparty VC syndicates when the prospective syndicate is characterized by greater status heterogeneity.

Hypothesis 3b. Prior tie density will have a stronger positive impact on the formation of multiparty VC syndicates when the prospective syndicate is characterized by greater status-based subgroups.

Industry Popularity

As targets for venture capital investment, industry sectors vary significantly in terms of how popular (or “hot”) they may be. These variations manifest themselves across industry sectors as well as time periods. For example, according to historical trend data in the PwC MoneyTree Report¹², in 2001, the number of early-/seed-stage VC investment deals was 431 in the software sector and 157 in the biotechnology sector. Over time, however, the number of early-/seed-stage deals in software declined from 431 in 2001 to 388 in 2008, while the number of similar stage deals in biotechnology almost doubled from 157 in 2001 to 311 in 2008.

The popularity of industry sectors can wax or wane for three intertwined reasons: technological developments, market crowding, and herd behavior (Agarwal & Tripsas, 2008; Banerjee, 1992). The launch of the commercial Internet around 1994 enabled e-commerce and spawned a large number of startups focused on this sector. Apple’s launch of the iPhone in 2007 and the App Store in 2008 enabled mobile games and spawned the rapid emergence of hundreds of

mobile gaming companies. Similarly, the completion of the Human Genome Project in 2003 has been a major driver behind the growth of new ventures in the biotechnology sector. Beyond the initial technological thrust, however, herd behavior tends to take over as new founders observe, learn from, and emulate the behavior of the pioneers (Banerjee, 1992). As accelerating cascades of information, fueled by the media, propel the herd behavior forward, optimism can start to overcome prudence (Sorenson & Stuart, 2008). Too many copycat ventures are launched, VCs compete to sign deals in a hurry, and valuations start going up at an accelerating pace (Valliere & Peterson, 2004). Eventually, the bubble bursts and slower market growth or even a decline (relative to the number of previous participants) may dampen the industry sector’s popularity, until new technological developments trigger a new cycle.

VCs face a particularly acute challenge during periods of high industry popularity. Hot industries are often perceived as particularly attractive because they are undergoing a period of growth and disruption (Eisenhardt & Schoonhoven, 1990); yet, such change may also mean that VCs face higher market uncertainty (i.e., external uncertainty) rooted in industry conditions and shared across a set of firms (Beckman, Haunschild, & Phillips, 2004). This is illustrated in Valliere and Peterson’s (2004: 1) interviews with 57 VCs who made early-stage investments during the 1998–2001 period, a period they characterized as “investors competing intensively in an unfamiliar sector with unknown success criteria.” Firms facing higher market uncertainty tend to engage in “satisficing” search behaviors (March, 1988) and partner with those firms they know best, thus making syndicates with higher tie density more attractive in hot industries.

Popular industries may also be characterized by more intense competition among investors, all of whom need to make investment decisions more expeditiously. These conditions make it more likely that prospective syndicates that actually become a reality will be those that can achieve more effective coordination at a faster pace. As one VC we interviewed noted:

“One of my big fears is ‘FOMO,’ or fear of missing out. Especially when the market is hot. In any particular space, you may have five startups instead of just two. I don’t have the luxury of time to carry out proper due diligence on all of them. It’s a real challenge to figure out which one to bet on. Yet, decide I must. So, what do I do? Bank on my relationships.”

We heard similar sentiments from a number of VCs. Fear of missing out, or “FOMO” as it is referred

¹² PwC MoneyTree Report. Retrieved November 28, 2014, from <https://www.pwc.com/us/en/technology/moneytree/explorer.html#/>.

to in the field, may be regarded as a form of loss aversion (Brehm, 1966; Tversky & Kahneman, 1981). Consistent with Eisenhardt and Schoonhoven's (1990) finding that ventures in high growth industries are generally more successful, VCs may conclude more popular industries offer more attractive deals. Faced with competition from other investors, they are likely to sense greater urgency in identifying the right ventures and making attractive investment offers. Popular investment industries are thus likely to be high-velocity environments requiring faster decision-making and syndication formation (Eisenhardt, 1989). As reflected in the above quote, when forming multi-party syndicates, a key way in which VCs may seek to accelerate such decisions is by relying more heavily on prior ties, which may address the coordination and cooperation challenges that otherwise accompany such industry popularity. More formally:

Hypothesis 4. Prior tie density will have a stronger positive impact on the formation of multi-party VC syndicates at higher levels of industry popularity.

METHODS

Data and Sample

We tested the hypotheses in this paper using data on VC investments in U.S.-based ventures during the period 1985–2008. The VentureXpert database constituted our primary data source. We constructed a sample of real syndicates along with matching hypotheticals (as discussed later) using the following procedures. First, we focused only on syndicates of a size of three or more formed at the time of first-round investments in a venture. This ensures all VC firms' prior relationships are comparable, in that all co-investors are new to the focal venture. Second, since investing in "mature" ventures is more similar to private equity than venture capital investments, and since deal structures and collaboration may differ across the two, we limited the study to ventures less than or equal to ten years old at the time of the first-round investment. Third, we included only first-round investments at the "Startup/Seed," "Early," or "Expansion" stage, since ventures at an earlier stage are likely to face greater uncertainty and thus require greater collaboration among the co-investing partners in the syndicate. Fourth, because the cross-border investing behavior of

foreign investors may differ from that of U.S.-based VC firms, we included only those first-round syndicates involving no foreign VC firms. Fifth, since angels are generally less structured in doing due diligence and often invest far less capital, they were not included in the analysis. During the period 1985–2008, there were 2,570 U.S.-based ventures with first-round syndicates formed by three or more identifiable VC firms;¹³ 2,134 of these were less than or equal to ten years old. Within this subset, 1,979 were at the "Startup/Seed," "Early," or "Expansion" stage at the time of the first-round investment. Of these, 1,588 had only U.S.-based VC investors. But for one additional data screen (discussed later in this section), it is this set of ventures that constitutes the basis for our empirical analysis.

Consistent with prior tie formation research (Hallen, 2008; Jensen, 2003; Sorenson & Stuart, 2008), we adopted a choice-based sampling approach. As detailed in the next paragraph, we generated matched hypothetical "groups" (rather than "dyads") for each venture. These hypothetical syndicates are those that could have been formed but were not. Including only a random sample of hypothetical syndicates, rather than all potential syndicates, has two advantages. First, it is computationally more tractable. Second, it helps to reduce autocorrelation among the analyzed groups, since each VC firm would have appeared in the sample repeatedly if we had included all possible syndicates. Moreover, these benefits come with relatively few costs, as prior literature has shown that using a matching ratio equal to 1:5 provides estimates highly consistent with including all non-realized outcomes (King & Zeng, 2001).

To construct the random sample of hypothetical syndicates for each real syndicate, we first generated a potential VC pool for each venture by identifying all U.S. VCs investing in the same industry, in the same state, in the same year, and in the same quarter. These criteria ensure that VC firms included in a potential pool were at high risk of joining a particular syndicate for the focal venture. From the potential VC pool for each real investment, we

¹³ Some VC firms involved in the syndicates were identified as undisclosed firms, and so we dropped all cases with any undisclosed VC firms. While generally only one or two VC firms in a syndicate round were not disclosed, it is theoretically important to drop such cases as the unobserved intragroup ties of the undisclosed VC firms may influence the focal constructs.

generated matched samples by randomly selecting VC firms to form hypothetical syndicates of the same size.¹⁴ Using a random selection in such choice-based sampling ensures that, when we select on the dependent variable, we would not select differently on the independent variables for the real and matched samples. Based on the random selection, a hypothetical syndicate could be completely different from the real syndicate group, or it could differ by just one or two VC firms, etc.

For each real case, five hypothetical syndicates were randomly selected. In a small number of cases, the VC firms in the potential pool for a specific venture were exactly the same as those involved in the actual syndicate and no extra VC firms were available. In these cases, it was not possible to create any hypothetical cases for these syndicates and they were eliminated from the sample. Further, about 4% of the real cases had less than five hypotheticals. We excluded these cases also from our sample. Thus, our final sample consisted of 1,336 U.S.-based ventures with three or more U.S.-based VC firms making first-round investments in the venture during the period 1985–2008. For these ventures, our dataset includes real syndicates and matching hypothetical syndicates. Our focus in this paper is predominantly on group-level constructs that are difficult to measure and test at the dyad level (e.g., prior tie density, network faultlines, status-based subgroups), and such a group-level perspective provides a less

biased, more straightforward, and easier-to-interpret analysis of multi-VC syndicate formation.

Consistent with previous research based on choice-based samples (e.g., Sorenson & Stuart, 2008), we used a conditional logit model grouped on ventures to test the hypotheses. The level of analysis was the syndicate. Using choice-based sampling and conditional logit allowed us to uncover the probability of a group of VC firms forming a syndicate for a certain venture. The hypothesized relationships were shown as interaction terms between prior tie density and other independent variables. This design permitted us to test for the joint effect of prior tie density and other contextual attributes. While controlling for venture and round-level unobservable factors (e.g., perceived venture quality), one tradeoff of the conditional logit analysis approach is that main effects cannot be estimated for variables that vary at the venture and round-level but not at the syndicate level.¹⁵ Finally, we used mean-centering for the main effect variables in the regressions, as doing so makes it easier to interpret the results.

Measures

Syndicate formation. The dependent variable in our analysis was the formation of a specific syndicate, with “1” referring to a syndicate formed in reality and “0” referring to a hypothetical syndicate that could have been formed but was not.

Prior tie density. Prior tie density was measured as the ratio of preexisting dyadic ties to the total number of possible dyadic ties within the prospective syndicate. Consistent with prior research on syndication in the VC context (Hallen, 2008; Hochberg et al., 2007; Sorenson & Stuart, 2001, 2008), a prior tie was deemed to have been created when two VC firms had co-invested at least once in the same round for the same venture during the five years prior to the new focal investment. Prior tie density is a continuous variable ranging between 0 and 1. For instance, assume that a potential syndicate involves three firms: A, B, and C. Firms A and B have a prior tie. So have firms A and C. However, firms B and C are new to each other (i.e., without any

¹⁴ Since a hypothetical syndicate is generated based on a random combination of potential VCs in the pool, if the size of the hypothetical syndicates for a venture is not controlled, hypothetical syndicates with larger size will have a greater number of combinations and account for a higher proportion in the hypothetical syndicate distribution for the venture. This will make it difficult to select a number of representative hypothetical syndicates for the venture, since a random selection will result in an overrepresentation of larger hypothetical syndicates in the matched sample. Moreover, although syndicate size is the same for the counterfactuals and the real syndicate for each venture, we were able to test for the moderating effect of syndicate size using within-group variation. As we hypothesized in Hypothesis 1, the formation of a syndicate is likely to be a function of the interaction between tie density and syndicate size. Due to these considerations, we only generated hypothetical syndicates whose sizes were equal to that of the real syndicate for the venture. Moreover, this is consistent with the approach adopted by Mitsuhashi and Greve (2009), who generated matching alliances of the same size by randomly replacing all members in a realized alliance.

¹⁵ As shown in previous research (Sorenson & Stuart, 2008), the fact that these variables drop out in the regression models does not affect our interpretation of the results. The focus here is on how the impact of prior tie density on syndicate formation may change depending on other variables, such as syndicate size and industry popularity, rather than the other way around.

prior co-investing history in the prior five years). The prior tie density here will be 2/3, since two out of three dyads have a prior collaboration during the five-year window.

Syndicate size. We measured syndicate size using the total number of VC firms investing together to form a syndicate.

Network faultlines. Prior literature has used measures of within-group dispersion to estimate subgroup strength and risk of faultlines (Gibson & Vermeulen, 2003; Heidl et al., 2014). Following these studies, we measured network faultlines using standard deviation of tie strength among all dyads in a prospective syndicate, and used log transformation to correct for the skewness of this variable. Since network faultlines has a high correlation ($r \geq .70$) with prior tie density and average tie strength, we orthogonalized network faultlines against prior tie density and average tie strength to ensure that our measure of network fault line risks captures the level of such risks above and beyond that accounted for by prior tie density and average tie strength; this was done using the Gram–Schmidt procedure and the *orthog* command in Stata software (StataCorp).

Status heterogeneity. Following prior work in various contexts including venture capital investments, we captured VC status using centrality in the co-investment network (Gulati & Gargiulo, 1999; Podolny, 2001). We first calculated firm status using Bonacich's measure of centrality (Bonacich, 1987):

$$c_i(\alpha, \beta) = \sum_{j=1}^n (\alpha + \beta c_j) R_{ij}$$

where c_i is the centrality score of firm i , R_{ij} measures the relationship between firm i and j , α scales the measure, and β weights the centrality of all firm i 's alters. Consistent with prior research (Sorenson & Stuart, 2001), we set β equal to three-quarters of the reciprocal of the largest eigenvalue. Following Gulati and Gargiulo (1999), we then calculated status similarity within each pair of VCs in a potential syndicate by dividing the minimum status by the maximum status within a pair. We then converted this to a group-level measure of status heterogeneity, calculated as 1 minus the average of the pairwise measures of status similarity in the syndicate.

Status-based subgroups. Following Gibson and Vermeulen (2003), we measured status-based subgroups by computing the standard deviation of status similarity among all dyads within the prospective syndicate; this approach is also broadly consistent with Heidl et al.'s (2014) approach to capturing

network faultlines and our measure of network fault line risks.

Industry popularity. Industry popularity was measured as the total number of ventures at the "Startup/Seed," "Early," or "Expansion" stage receiving first-time VC funding in the same industry segment as the focal venture in the prior year. Consistent with Sorenson and Stuart (2008), we identified ten industry segments based on the VentureXpert database.¹⁶ We used the log transformation of this variable to adjust for skewness and to capture the nonlinear impact of increasing industry popularity. As described in the robustness tests section, below, we also ran analyses using alternative approaches to capturing this construct, and obtained broadly similar results.

Control variables. We also included a number of control variables consistent with prior studies of dyad-level VC syndication. In some cases, a group of co-investors may include not only independent VC firms but also other types of VC firms such as corporate VCs, bank-affiliated VCs, and so forth (Hallen, Katila, & Rosenberger, 2014). Based on the salience of homophily and concerns about the motivations of other types of investors, VC firms of similar type may be more likely to come together in a syndicate. We controlled for this effect by including a measure for *VC type heterogeneity*. Similar to the approach used by Joshi and Roh (2009) and Teachman (1980), we calculated VC type heterogeneity by using an entropy-based measure, $\sum P_i \ln(1/P_i)$, where P_i is the proportion of a specific VC type in a syndicate.¹⁷

Prior dyad-level research has found that two firms are more likely to form a tie when there is an indirect tie between them (Gulati, 1995; Gulati & Gargiulo, 1999; Hallen, 2008; Sorenson & Stuart, 2001). An indirect tie between two firms can be said to exist when they are separated by a geodesic distance of

¹⁶ The ten industry segments classified by Venture Economics Industry Codes cover biotechnology, communications and media, computer hardware, computer software and services, consumer related, industrial/energy, internet specific, medical/health, semiconductors, and other products.

¹⁷ The different types involved in our sample are private equity firm (74.2%), corporate PE/venture (7.7%), bank affiliated (9.6%), private equity advisor (1.7%), SBIC (1.6%), incubator program (0.8%), insurance firm affiliate (0.8%), endowment, foundation or pension fund (0.8%), service provider (0.8%), investment management firm (0.6%), government affiliated program (0.4%), angel group (0.4%), university program (0.3%), and other (0.3%). On average, 2.6 different types of VC investors are involved in a multifirm syndicate.

two. Thus, if two firms have not invested together in the prior five years, but both of them have co-invested with a common third party, then these two firms have an indirect tie. In this paper, due to the importance of the syndicate context, we differentiated between cases in which the indirect tie was formed via common *insiders* versus common *outsiders*. To calculate *internal indirect tie density*, we first identified whether there was an indirect tie between two partners going through another VC in the same prospective syndicate. We then calculated the proportion of pairs in the potential syndicate that were connected by at least one internal indirect tie. We used a similar approach to compute *external indirect tie density*, calculating the proportion of pairs in the potential syndicate that were connected by at least one external indirect tie through a firm outside of the prospective syndicate.

Because VC firms closer in geographic distance are more likely to partner with one another (Sorenson & Stuart, 2008) and prior tie density might facilitate syndicate formation among partners far from one another, we controlled for the average *geographic distance* among potential syndicate partners and its interaction with prior tie density. To measure geographic distance, we used zip codes to collect data on the corresponding latitudes and longitudes of VC firm offices and calculated the distance between two VC firms using spherical geometry. To compute geographic distance at the syndicate level, we then calculated the average of dyadic geographic distance and did a log transformation to capture the nonlinear impact of geographic distance. About 2% of the syndicates in our sample had missing values on geographic locations of VC firms. In these cases, we used only those VC firms with available geographic information to calculate geographic distance among the syndicate members.¹⁸

Similarly, following Sorenson and Stuart (2008), we controlled for average *industry distance* among the potential syndicate partners and its interaction with prior tie density. To measure industry distance within the syndicate, we first computed the industry specialization of syndicate members across the ten Venture Economics Industry Code segments in the prior five years. Following Sorenson and Stuart (2008), we then calculated the squared deviation for each VC dyad across the ten industry segments, $\Sigma(P_{ik} - P_{jk})^2$, where P_k represents the percentage of

investment volume in the k th industry segment and i or j refers to VC syndicate members. Next, we computed the average of the squared deviation across all dyads in the syndicate and did a log transformation to correct for the skewness of this measure.

Previous dyadic literature has found that firms were more likely to partner with unfamiliar firms when the contextual uncertainty was low (e.g., Sorenson & Stuart, 2008). Since this effect may hold at the group level as well, we controlled for the effect of contextual uncertainty. Specifically, since the developmental stage of a venture may influence the uncertainty associated with the investment, we used *venture age* as a proxy for contextual uncertainty (Hallen, 2008). Younger ventures are likely to be associated with a higher level of contextual uncertainty. While the main effect of venture age is captured in the fixed effects of our conditional logit model, we included an interaction of venture age and prior tie density.

We also controlled for average tie strength among potential syndicate partners to better isolate the effect of network faultlines (which is also calculated based on underlying tie strength). Since the network structure of a syndicate influences which ties are even present, we constructed this measure as *relative tie strength*, measured as average tie strength (log transformed) orthogonalized relative to prior tie density using the Gram–Schmidt procedure and the *orthog* command in Stata. Relative tie strength thus captures tie strength in a syndicate above or below what would be expected given the underlying density of the syndicate.

Finally, to better disentangle status within a potential syndicate from our focal constructs capturing status disparities, we controlled for the *average status* of potential syndicate partners (logged to correct for skewness). Consistent with the prior measures, we measured status of each VC using Bonacich centrality in the co-investment network.

RESULTS

Table 1 presents the descriptive statistics for and zero-order correlations between all variables in our sample.¹⁹ Consistent with our baseline expectation, a positive and moderate ($r = .17$) correlation between

¹⁸ As a robustness check, we also retested our models by dropping syndicates with missing values on any VC's geographic location and found consistent results.

¹⁹ It should be noted that, reflecting our broader empirical design, this sample contains five hypothetical syndicates for each realized syndicate; thus, the correlations should be understood more as reflecting the broader structure of the VC syndication network than the characteristics of only the realized syndicates.

TABLE 1
Descriptive Statistics and Correlation Table

	Variables	Mean	SD	Min.	Max.	1	2	3	4	5
1	Syndicate formation	0.17	0.37	0.00	1.00	1.00				
2	VC type heterogeneity	0.45	0.40	0.00	1.79	-0.04*	1.00			
3	Internal indirect tie density	0.05	0.12	0.00	0.60	0.06*	-0.03*	1.00		
4	External indirect tie density	0.47	0.31	0.00	1.00	-0.15*	-0.04*	-0.13*	1.00	
5	Geographic distance	6.42	1.64	0.00	8.13	-0.13*	0.09*	0.00	0.07*	1.00
6	Industry distance	0.24	0.16	0.00	0.98	-0.03*	0.13*	-0.23*	-0.09*	0.05*
7	Venture age	1.59	1.98	0.00	10.00	0.00	0.01	-0.01	0.00	-0.02
8	Relative tie strength	-0.00	1.00	-3.46	6.89	0.09*	-0.06*	-0.01	0.00	-0.01
9	Average status	0.99	0.39	0.00	2.15	0.01	-0.11*	0.33*	0.12*	0.02*
10	Prior tie density	0.25	0.28	0.00	1.00	0.17*	-0.12*	0.50*	-0.40*	-0.10*
11	Syndicate size	3.59	0.99	3.00	11.00	0.00	0.19*	0.22*	-0.03*	0.12*
12	Network faultlines ^a	-0.00	1.00	-9.36	4.49	-0.04*	0.09*	0.20*	0.10*	0.07*
13	Status heterogeneity	0.67	0.20	0.02	1.00	-0.02	0.16*	-0.29*	-0.23*	0.03*
14	Status-based subgroups	0.21	0.10	0.00	0.49	-0.01	0.02*	0.04*	0.12*	0.05*
15	Industry popularity	4.76	0.99	1.79	6.95	0.00	0.07*	-0.18*	0.02*	0.02*

syndicate formation and prior tie density can be seen. Consistent with prior literature on interorganizational collaboration, a .57 correlation is here observable between prior tie density and average status—indicating, as expected, that high-status VCs (who generally have broader networks) are more likely to have previously collaborated with one another (Gulati & Gargiulo, 1999; Podolny, 1994).

The correlations in Table 1 also indicate that our regressions are unlikely to be biased by multicollinearity. Supporting this, all variance inflation factors (including all interaction terms) in our regression models were well below the traditional threshold of 10, and the mean variance inflation factor is 1.89. Similarly, the overall conditionality index is only 11.86, well below the traditional threshold of 30, further indicating that multicollinearity is unlikely to be an issue in our models.

Table 2 presents empirical tests of our hypotheses about syndicate formation. Since syndicates containing the same VC firms may cause autocorrelation biases, we followed Lincoln (1984) to control for this autocorrelation. We created a variable for each syndicate by computing the mean of the dependent variable across all other syndicates in the same year that include any VC firms in the focal syndicate, and included this variable in our group-level analyses. Model 1 contains only the control variables and the main effects of the theory variables. As expected, prior tie density has a strong positive effect on likelihood of syndicate formation, thereby confirming our baseline premise that prior tie density promotes the likelihood of future collaboration. Consistent with arguments that VCs may be cautious about

investing alongside different types of investors (e.g., corporate venture capital units, etc.) as such investors may invest for non-financial reasons and may even be competing with the ventures in which they invest (Hallen et al., 2014; Katila, Rosenberger, & Eisenhardt, 2008), we find that VC type heterogeneity negatively influences syndicate formation. Consistent with prior studies examining dyadic syndication amongst pairs of VCs (Sorenson & Stuart, 2008), the positive impact of prior tie density is stronger when prospective syndicate partners are geographically and industrially distant from each other, while venture age does not appear to alter the importance of prior tie density. Also consistent with expectations, relative tie strength has an additional positive effect on syndicate formation above and beyond prior tie density.

While our theory focuses on how various contextual factors alter the relative importance of tie density on syndicate formation, Model 1 also offers insights into the direct effects of many of these contextual factors. Although syndicate size and industry popularity are invariant across the realized and the hypothetical syndicates, and thus their direct effects cannot be estimated in our conditional logit models, we find a negative relationship between the risk of network faultlines and syndicate formation ($p = .047$). This is consistent with our arguments that the risk of network faultlines may deter the formation of a particular multiparty syndicate.

Model 1 also offers insight into the status dynamics of multiparty syndication. First, we find that potential syndicates with higher average status are less likely to form; this is consistent with the idea that

TABLE 1 (continued)

6	7	8	9	10	11	12	13	14	15
1.00									
0.01	1.00								
-0.06*	-0.04*	1.00							
-0.46*	-0.07*	0.17*	1.00						
-0.41*	-0.03*	0.00	0.57*	1.00					
0.03*	-0.02*	0.11*	-0.01	0.00	1.00				
0.00	0.02	0.00	0.08*	0.00	0.43*	1.00			
0.44*	0.03*	-0.02*	-0.41*	-0.51*	0.00	0.06*	1.00		
-0.03*	-0.01	0.13*	0.11*	0.01	0.24*	0.25*	-0.13*	1.00	
0.12*	-0.01	-0.09*	-0.13*	-0.26*	-0.04*	-0.05*	0.20*	-0.03*	1.00

* $p < .05$ (one-tailed)

^a *Network faultlines* is orthogonalized relative to prior tie density and average tie strength using the Gram–Schmidt procedure.

there are fewer high-status VCs relative to the entire population (Podolny, 2001), and thus high-status syndicates are also likely to be relatively infrequent. Whereas literature on the formation of network dyads has tended to emphasize the positive influence of status homophily, we find instead that status heterogeneity is positively associated with the formation of multi-VC syndicates ($p = .001$). Building on our fieldwork that indicated several positive benefits to status heterogeneity in multiparty syndicates (e.g., high-status VCs being able to obtain a higher percentage of equity), this empirical finding indicates that the benefits of status heterogeneity may, on average, outweigh the perceived drawbacks. Finally, we do not find a statistically significant direct effect for status-based subgroups. Overall, these results indicate that status dynamics in multiparty collaborations may differ sharply from the status dynamics of dyadic formation, a point we return to later in the Discussion section (below).

Models 2–7 contain the results of the tests of our hypotheses, with Models 2 to 6 adding interaction terms one at a time and Model 7 presenting the full model including all interactions. In Hypothesis 1, we argued that prior tie density would have a stronger positive impact on the formation of multiparty VC syndicates when the size of the potential syndicate is larger. In support of this argument, we find that the interaction of prior tie density and syndicate size is positive and highly significant ($p = .001$) in both the partial Model 2 and the full Model 7. Moreover, we find that this result is robust while controlling for

both the average Bonacich centrality (average status) of potential syndicate members, and, as detailed in the robustness tests, the average degree centrality of those potential members. These results suggest that greater tie density does become a more important enabling factor as the size of a syndicate grows. Based on Model 7, we find a one standard deviation increase in prior tie density from its mean yields a 66% greater likelihood of syndicate formation in a five-firm syndicate, as compared with a three-firm syndicate.²⁰

In Hypothesis 2, we argued that prior tie density would have a stronger positive impact on the formation of multiparty VC syndicates when the prospective syndicate suffers from a higher risk of network faultlines. Consistent with Heidl et al. (2014), in Model 1, we find that network faultlines has a negative and significant main effect ($p = .047$), indicating that network faultlines do appear to be a concern in the formation of multiparty VC syndicates. Contrary to Hypothesis 2, however, the interaction of prior tie density and network faultlines is not significant in either the partial Model 3 or the full Model 7. One possible interpretation is that, since network fault line risks are ultimately network based, such concerns may override the trust and familiarity normally provided by greater tie density. We return to this possibility in the Discussion.

²⁰ All effect magnitudes are calculated using the *margins* command in Stata, and setting all other variables to their mean value.

TABLE 2
Hypotheses Testing: Conditional Logit Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DV: Syndicate formation							
VC type heterogeneity	−0.20* (0.09)	−0.20* (0.09)	−0.20* (0.09)	−0.20* (0.09)	−0.20* (0.09)	−0.20* (0.09)	−0.21* (0.09)
Internal indirect tie density	−0.12 (0.31)	−0.20 (0.31)	−0.11 (0.32)	−0.20 (0.31)	−0.20 (0.31)	−0.15 (0.31)	−0.25 (0.33)
External indirect tie density	0.06 (0.18)	0.06 (0.18)	0.06 (0.18)	−0.08 (0.19)	0.04 (0.18)	0.03 (0.18)	−0.09 (0.19)
Geographic distance	−0.17*** (0.02)	−0.17*** (0.02)	−0.17*** (0.02)	−0.17*** (0.02)	−0.17*** (0.02)	−0.17*** (0.02)	−0.17*** (0.02)
Industrial distance	0.17 (0.29)	0.18 (0.29)	0.17 (0.29)	0.12 (0.29)	0.19 (0.29)	0.19 (0.29)	0.15 (0.29)
Prior tie density	3.92*** (0.28)	4.12*** (0.28)	3.90*** (0.29)	3.86*** (0.28)	3.92*** (0.28)	3.93*** (0.28)	3.98*** (0.30)
Prior tie density × Venture age	−0.07 (0.06)	−0.06 (0.06)	−0.07 (0.06)	−0.07 (0.06)	−0.07 (0.06)	−0.08 (0.06)	−0.07 (0.06)
Prior tie density × Geographic distance	0.20*** (0.05)	0.18*** (0.05)	0.20*** (0.05)	0.20*** (0.05)	0.20*** (0.05)	0.20*** (0.05)	0.17** (0.05)
Prior tie density × Industrial distance	1.99* (0.99)	2.03* (0.99)	1.97* (1.00)	1.37 [†] (1.04)	1.94* (0.99)	1.95* (0.99)	1.27 (1.05)
Relative tie strength	0.37*** (0.03)	0.36*** (0.03)	0.37*** (0.03)	0.37*** (0.03)	0.36*** (0.03)	0.37*** (0.03)	0.36*** (0.03)
Average status	−1.41*** (0.14)	−1.41*** (0.14)	−1.41*** (0.14)	−1.37*** (0.14)	−1.38*** (0.14)	−1.44*** (0.14)	−1.40*** (0.15)
Network faultlines ^a	−0.06* (0.04)	−0.07* (0.04)	−0.06 (0.05)	−0.09* (0.04)	−0.07* (0.04)	−0.08* (0.04)	−0.07 (0.06)
Status heterogeneity	1.28*** (0.26)	1.30*** (0.26)	1.28*** (0.26)	1.06*** (0.28)	1.11*** (0.27)	1.24*** (0.26)	0.97*** (0.29)
Status-based subgroups	0.17 (0.35)	0.16 (0.35)	0.17 (0.35)	−0.15 (0.39)	0.12 (0.36)	0.13 (0.36)	−0.19 (0.39)
Autocorrelation control	−3.46*** (0.38)	−3.44*** (0.37)	−3.46*** (0.38)	−3.44*** (0.38)	−3.45*** (0.38)	−3.44*** (0.38)	−3.40*** (0.37)
H1: Prior tie density × Syndicate size		0.68*** (0.18)					0.73*** (0.19)
H2: Prior tie density × Network faultlines			−0.02 (0.10)				−0.11 (0.11)
H3a: Prior tie density × Status heterogeneity				1.55* (0.76)			1.42* (0.77)
H3b: Prior tie density × Status-based subgroups					2.77* (1.27)		1.92 [†] (1.31)
H4: Prior tie density × Industry popularity						0.43** (0.13)	0.43** (0.13)
N	8016	8016	8016	8016	8016	8016	8016
χ ²	771.93	786.69	771.96	776.00	776.74	782.31	805.39
Log likelihood	−2007.83	−2000.44	−2007.81	−2005.79	−2005.42	−2002.64	−1991.09
Pseudo R ²	0.16	0.16	0.16	0.16	0.16	0.16	0.17

Note: Standard errors in parentheses.

[†] $p < .10$

* $p < .05$

** $p < .01$

*** $p < .001$ (one-tailed)

^a Network faultlines was orthogonalized relative to prior tie density and average tie strength using the Gram–Schmidt procedure.

In Hypothesis 3a, we argued that prior tie density would have a stronger positive impact on the formation of multiparty VC syndicates when the prospective syndicate is characterized by greater status heterogeneity. Supporting Hypothesis 3a, the interaction of prior tie density and status heterogeneity is positive and significant in both the partial Model 4 ($p = .022$) and the full Model 7 ($p = .033$). This result is consistent with our fieldwork and arguments, which suggested that prior tie density may reduce the hazards that may accompany status heterogeneity in multiparty VC syndicates, thereby enabling the syndicate members to take advantage of the many positive benefits resulting from greater status heterogeneity (e.g., the higher-status VCs get to keep more equity while the lower-status VCs benefit from status accrual). Based on the results of Model 7 in Table 2, we find that a one standard deviation increase in prior tie density from its mean yields a 23% greater likelihood of syndicate formation when status heterogeneity is at one standard deviation above the mean, as compared with when it is at the mean level.

Paralleling our arguments for status heterogeneity, Hypothesis 3b considered a different form of status disparities: status-based subgroups. This construct (and our measure) captures the extent to which a potential syndicate has subgroups that are internally similar in firm status, but where different subgroups have very different status levels. Consistent with Hypothesis 3b, we find a positive and significant interaction of prior tie density and status-based subgroups ($p = .015$) in the partial Model 5, although the interaction is only near-significant ($p = .071$) in the full Model 7. As detailed later in our robustness tests, however, we find that the interaction is positive and of higher significance ($p < .05$) under certain conditions (e.g., younger ventures, higher-quality ventures, midsize rounds), thereby offering nuanced partial support to Hypothesis 3b. Based on the results of Model 7 in Table 2, we find that a one standard deviation increase in prior tie density from its mean yields a 6% greater likelihood of syndicate formation where status-based subgroups is at one standard deviation above the mean, as compared with when it is at the mean level.

Finally, we theorized in Hypothesis 4 that prior tie density would have a stronger positive impact on the formation of multiparty VC syndicates at higher levels of industry popularity. In support of this argument, we find a positive and significant interaction between prior tie density and industry

popularity in both the partial Model 6 ($p = .001$) and the full Model 7 ($p = .001$). The results of Model 7 in Table 2 suggest that one standard deviation increase in prior tie density from its mean yields a 15% greater likelihood of syndicate formation when industry popularity is at one standard deviation above the mean, as compared with when it is at the mean level.

Robustness Tests

We also ran a number of additional robustness tests to explore the sensitivity of the results to our measurements and sampling choices. Due to space considerations, only part of the results are reported in Table 3. As shown in Table 3, we dropped the 1% of observations with the highest residuals on syndicate formation (a total of 14 rounds) in Model 8 to ensure that our results were not being driven by possible outliers. We replaced industry popularity with change in industry popularity²¹ (measured as the number of ventures in a focal year within the focal venture's industry minus the average number across the preceding three years) in Model 9 to ensure that our measure of industry popularity was not sensitive to arbitrary choices of industry boundaries. We also controlled for round pre-money valuation (a proxy for venture quality, according to Hochberg et al., 2007) in Model 10 by including its interaction with prior tie density for the 500 rounds where data are available. In Model 11, we controlled for average investor experience,²² which was omitted from the main analysis due to its high correlation with average VC status ($r = .67$). Model 12 controlled for average investor degree centrality by including its orthogonalized measure relative to average status (average

²¹ We are thankful to an anonymous reviewer for suggesting this robustness test. Since the choice of three years can be arbitrary, we also used averages of the previous two, four, or five years when calculating change in industry popularity, and found consistent results. To rule out other alternative explanations, we also conducted the analyses using number of ventures in all rounds (rather than first round only), number of first-round ventures in the focal year (rather than prior year), number of VC firms investing in the first round, and number of VC firms newly entered. All of these alternative measures were calculated in the same industry of the focal venture and limited to the ventures at "Startup/Seed," "Early," or "Expansion" stage or VC firms invested at these stages, and yield robust results.

²² We used years of investment experience instead of age as it is better generalized to corporate venture capital units, where the parent corporation may be quite old but the corporate venture capital unit may be relatively new.

TABLE 3
Robustness Tests of Measurements and Sampling Choices

	(8) Drop 1% Outliers	(9) Change in Industry Popularity	(10) Control for Pre- Money Valuation	(11) Control for Average Investor Experience	(12) Control for Average Investor Degree	(13) Excluding Solo Investors	(14) Midsize Rounds (3–4 Investors)
DV: Syndicate formation							
VC type	−0.21*	−0.19*	−0.23 [†]	−0.20*	−0.20*	−0.28**	−0.25*
heterogeneity	(0.10)	(0.09)	(0.16)	(0.09)	(0.09)	(0.10)	(0.10)
Internal indirect	−0.29	−0.17	−0.63	−0.24	−0.23	0.09	−0.41
tie density	(0.33)	(0.33)	(0.58)	(0.33)	(0.33)	(0.29)	(0.36)
External indirect	−0.10	−0.12	0.16	−0.09	−0.13	−0.13	−0.01
tie density	(0.19)	(0.19)	(0.32)	(0.19)	(0.19)	(0.16)	(0.19)
Geographic	−0.18***	−0.17***	−0.19***	−0.17***	−0.17***	−0.16***	−0.16***
distance	(0.02)	(0.02)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)
Industry distance	0.15	0.14	0.38	0.14	0.17	0.16	0.18
	(0.30)	(0.29)	(0.49)	(0.29)	(0.29)	(0.30)	(0.30)
Prior tie density	4.17***	3.95***	4.28***	3.98***	3.97***	3.74***	3.98***
	(0.31)	(0.30)	(0.72)	(0.30)	(0.30)	(0.35)	(0.34)
Prior tie density	−0.07	−0.05	−0.18 [†]	−0.07	−0.07	−0.12*	−0.05
× Venture age	(0.06)	(0.06)	(0.12)	(0.06)	(0.06)	(0.06)	(0.06)
Prior tie density	0.17***	0.17***	0.19*	0.17***	0.18***	0.11*	0.16***
× Geographic	(0.05)	(0.05)	(0.09)	(0.05)	(0.05)	(0.05)	(0.05)
distance							
Prior tie density	1.28	1.10	1.46	1.21	1.13	0.86	1.13
× Industry	(1.07)	(1.05)	(1.84)	(1.05)	(1.06)	(1.03)	(1.08)
distance							
Relative tie	0.39***	0.35***	0.34***	0.36***	0.36***	0.37***	0.33***
strength	(0.03)	(0.03)	(0.06)	(0.03)	(0.03)	(0.03)	(0.03)
Average status	−1.45***	−1.40***	−1.25***	−1.30***	−1.40***	−1.45***	−1.33***
	(0.15)	(0.15)	(0.24)	(0.16)	(0.15)	(0.15)	(0.15)
Network	−0.08 [†]	−0.06	−0.11	−0.07	−0.07	−0.15*	−0.07
faultlines ^a	(0.06)	(0.06)	(0.09)	(0.06)	(0.06)	(0.07)	(0.07)
Status	1.01***	0.97***	0.93*	0.94***	0.92***	1.06***	0.91**
heterogeneity	(0.29)	(0.29)	(0.51)	(0.29)	(0.29)	(0.28)	(0.29)
Status-based	−0.17	−0.22	0.09	−0.17	−0.22	0.10	−0.28
subgroups	(0.39)	(0.39)	(0.67)	(0.39)	(0.39)	(0.39)	(0.39)
Autocorrelation	−3.88***	−3.43***	−4.09***	−3.39***	−3.41***	−4.78***	−3.00***
control	(0.39)	(0.37)	(0.66)	(0.37)	(0.38)	(0.40)	(0.38)
H1: Prior tie	0.68***	0.74***	0.68*	0.73***	0.73***	0.60**	0.77*
density ×	(0.19)	(0.19)	(0.33)	(0.19)	(0.19)	(0.20)	(0.35)
Syndicate size							
H2: Prior tie	−0.09	−0.12	−0.03	−0.11	−0.11	0.03	−0.10
density ×	(0.11)	(0.11)	(0.18)	(0.11)	(0.11)	(0.11)	(0.12)
Network							
faultlines ^a							
H3a: Prior tie	1.42*	1.46*	1.35	1.39*	1.39*	1.52*	1.52*
density ×	(0.78)	(0.77)	(1.36)	(0.77)	(0.77)	(0.76)	(0.78)
Status							
heterogeneity							
H3b: Prior tie	2.22*	1.97 [†]	3.98*	1.90 [†]	2.04 [†]	−0.46	2.48*
density ×	(1.33)	(1.31)	(2.35)	(1.31)	(1.31)	(1.29)	(1.34)
Status-based							
subgroups							
H4: Prior tie	0.47***	0.00***	0.45*	0.43***	0.43***	0.54***	0.43**
density ×	(0.14)	(0.00)	(0.24)	(0.13)	(0.14)	(0.14)	(0.14)
Industry							
popularity ^b							

TABLE 3
(Continued)

	(8) Drop 1% Outliers	(9) Change in Industry Popularity	(10) Control for Pre- Money Valuation	(11) Control for Average Investor Experience	(12) Control for Average Investor Degree	(13) Excluding Solo Investors	(14) Midsize Rounds (3–4 Investors)
Prior tie density × Pre-money valuation			0.15 (0.28)				
Average investor experience				−0.01 [†] (0.01)			
Average investor degree ^c					0.06 (0.05)		
N	7932	8016	3000	8016	8016	7716	6900
χ ²	869.17	812.64	348.71	807.18	806.70	800.53	680.18
Log likelihood	−1934.12	−1987.47	−721.52	−1990.20	−1990.44	−1903.94	−1720.43
Pseudo R ²	0.18	0.17	0.19	0.17	0.17	0.17	0.17

Note: Standard errors in parentheses.

^a *Network faultlines* was orthogonalized relative to prior tie density and average tie strength.

^b In Model 9, the coefficient for Prior tie density × Industry popularity (using an alternative measure—change in industry popularity) is 0.0022, with standard error of 0.0005. This coefficient is significant at $p < .001$ level (one tailed).

^c *Average investor degree* was orthogonalized relative to average status (which is measured as Bonacich centrality). All orthogonalizations were done using the Gram–Schmidt procedure.

[†] $p < .10$

* $p < .05$

** $p < .01$

*** $p < .001$ (one-tailed)

Bonacich centrality), due to the high correlation ($r = .93$) between the two types of centrality measures. Because the interaction effect of industry popularity and prior tie density can be confounded by the presence of a larger number of new and, therefore, unconnected VCs in a more popular industry, we reconstructed our VC pool to include only those VC firms that had syndicated (i.e., excluding solo investors) in the same industry, state, year, and quarter. Note that these tighter restrictions reduce the sample size somewhat, as shown in Model 13. To verify that our results (for syndicate size and other variables) are not driven by relatively infrequent rounds containing a large number of investors, we also ran an analysis restricting our sample to investment rounds characterized by midsize syndicates (i.e., containing only three or four investors), with the results shown in Model 14.

Overall, the results in Table 3 suggest that our findings are robust to these alternative measurements and sampling specifications. As with Table 2, we consistently find support for the interactions of prior tie density and syndicate size (Hypothesis 1) and prior tie density and industry popularity (Hypothesis 4). Also consistent with Table 2, we do not find statistically significant support for Hypothesis 2

(i.e., the interaction of prior tie density and network faultlines) across any of these robustness tests. The interaction of prior tie density and status heterogeneity (Hypothesis 3a) is supported in six of these seven robustness tests. The exception is in Model 10, which includes the interaction of prior tie density and pre-money valuation—a difference likely due to either a much smaller sample size (roughly 37% of the original sample) or the possibly higher quality of ventures in this subsample.²³ The interaction of prior tie density and status-based subgroups (Hypothesis 3b) is positive and significant at the $p < .05$ level in three of the robustness tests, near significant ($p < .10$) in three of the robustness tests, and non-significant in one of the robustness tests. As a whole, these robustness tests offer further support for most of the findings in Table 2, except that the support of Hypothesis 3b only holds under certain circumstances, such as higher-quality ventures, midsize syndicates, and (as detailed below) younger ventures.

²³ Since the data on pre-money valuation are often revealed retrospectively from IPO disclosures when high-quality ventures do an IPO later on, it is very likely that these rounds reflect higher-quality ventures.

In addition to the robustness tests reported in Table 3, we also ran a number of other robustness tests, as follows:

- (1) We ran an analysis on an alternative sample, including only three hypothetical syndicates for one real syndicate.
- (2) We ran analyses using three-year syndication networks to construct the measures, although our choice of five-year syndication networks follows from prior literature (Hallen, 2008; Hochberg et al., 2007).
- (3) We also explored restricting the analysis to younger rounds in which ventures were less than or equal to five years in age.
- (4) Finally, we explored sensitivity to keeping the “lead” VC (i.e., who participated in the first-round investment and also invested in all subsequent rounds) constant across the matched hypothetical cases where such a lead could be clearly identified, even though our fieldwork indicated that many rounds did not have a clear lead VC.

These robustness tests also yield consistent results, except that Hypothesis 3a becomes near significant ($p < .10$) in tests (2)²⁴ and (3), in which Hypothesis 3b is significant ($p < .05$).

Overall, these robustness tests offer further support for our main analyses, as well as the insight that the positive interaction between prior tie density and status-based subgroups (Hypothesis 3b) is supported for ventures of higher quality, with midsized syndicates, and at a younger age.

DISCUSSION

In this paper, we have sought to explore the formation of multiparty collaboration in the context of first-round VC syndicates—a context that exhibits substantial multiparty collaboration but is not often examined through the multiparty lens. Using a rich quantitative sample and illustrative qualitative evidence, we have argued and shown that multiparty syndication is influenced by a number of *group-level* constructs and logics that would have no meaning at the dyadic level. As a result, the formation of multiparty syndicates cannot be viewed as a mere extension of dyadic constructs and logics. Collectively, our theory and findings offer several important contributions.

Implications for VC Syndication Literature

A primary contribution of our theory and empirical findings is a more complete perspective on how VC firms form syndicates, highlighting the importance of group-level constructs on multiparty syndicate formation. Our core argument is that, while providing syndicates with many benefits, group-level constructs or environmental factors may introduce a variety of cooperation and coordination challenges that amplify the importance of prior tie density. For instance, while larger syndicates may offer greater advice and access to broader networks, larger syndicates may be more difficult to coordinate and have a higher risk of at least one VC engaging in undesired behavior, such as pushing for an earlier or later exit than desired by the other investors. Our empirical findings indicate that the familiarity, trust, and closure provided by greater tie density help alleviate many of these concerns, making group-level attributes like larger groups or status disparities more desirable. Taken as a whole, these findings indicate that the formation of multiparty VC syndicates is not a simple deductive elaboration of dyad formation logics; rather, it involves different constructs and logics.

We also show a number of ways in which status dynamics differ across the formation of dyads versus multiparty VC syndicates. Prior literature on both VC syndication and other dyadic collaboration has often emphasized the presence of status homophily (e.g., McPherson et al., 2001; Podolny, 1994). While similarly experienced VCs (who are generally of similar status) have been shown to generally syndicate first rounds together (Lerner, 1994), we find in both our interviews with VCs and quantitative analyses that multiparty first-round VC syndicates actually tend to exhibit status disparities—especially when prior tie density is higher. This is consistent with Hochberg and colleagues (2015), who found little evidence for similarity based matching for VC syndication. Moreover, we articulate and show that there are actually two distinct types of status disparities in multiparty syndicates, both of which are positively associated with syndicate formation in denser syndicates: status heterogeneity (wherein a syndicate simply contains VCs of different status levels) and status-based subgroups (primarily influential in dense rounds where the venture is younger, of higher quality, and involving only three to four VCs). Explaining this dynamic, our qualitative fieldwork indicates that status disparities may often be beneficial in allowing higher-status VCs to

²⁴ We interpret these results in test (2) with some caution, as these models omit the influence of older albeit still relevant ties (i.e., ties occurring four or five years earlier).

make larger investments for a greater percentage of equity while still including a larger number of investors. More broadly, such theory and findings indicate that status disparities may often play a key role in facilitating an agreeable distribution of rewards in multiparty syndicates.

Finally, we also contribute to the VC literature by offering a detailed understanding of the VC syndication process. While prior literature has often portrayed a process whereby the entrepreneur reaches agreement with a lead VC who then invites and coordinates the investments of other VCs, our fieldwork indicated that this perspective may not always be accurate. Instead, we observed that VC syndication should be regarded as arising from mutual agreement among an evolving set of potential partners—for instance, a VC may introduce the entrepreneur to other VCs with whom they may have discussions, or entrepreneurs may bring other potential investors to already-interested VCs. A key implication of this dynamic is that each VC generally conducts at least some of its own diligence about both the venture and any syndication partners, and that syndicate formation reflects independent acceptance by each interested party. For our research, this mutual agreement perspective highlights the importance of considering all potential syndicate members when studying the formation of multiparty syndicates. For future research, this mutual agreement suggests that it may be insightful to study how entrepreneurs achieve such mutual agreement, the importance of the order in which different VCs become involved, or how VCs manage conflict that emerges at the time of syndicate formation.

Contributions to Multiparty Collaboration Literature

We also contribute to the broader network formation literature by explicating how group-level formation is not a simple deductive elaboration of dynamics at the dyadic level. As discussed above, we show that the formation of multiparty collaborations often depends on mutual agreement among all involved parties, thus requiring constructs, theoretical logics, and empirical methods that all account for the group-level nature of such collaborations. Moreover, while we have demonstrated the importance of a variety of group-level constructs that focus on the network and status structures of the group, we also recognize that other group-level factors may also influence such formation and that this is an interesting avenue for future research.

This paper also contributes by extending our understanding of faultlines in multiparty collaborations. Building on Heidl et al. (2014), who studied the effect of faultlines on multipartner alliance dissolution, we show that network faultlines also undermine the formation of multiparty VC syndicates. Unexpectedly, however, we did not find that greater tie density dampened the negative effects of network faultlines on syndication formation. This suggests network faultlines may be a persistent and hard-to-overcome issue in formed syndicates. Along with Heidl et al. (2014), our study thus reinforces that future research should factor in fault line risk in analyzing the formation, functioning, or dissolution of multifirm partnerships.

As we have focused on only a single context in the form of multiparty VC syndicates, one obvious question is the extent to which our theory and findings are likely to hold across other multiparty collaboration contexts such as multiparty alliances. One key assumption and boundary condition of our arguments is that such multiparty alliance formation depends on the mutual agreement of all involved parties—accordingly, if there is a single coordinating party and each member is concerned primarily with the lead coordinator but not other members, then we would expect the influence of constructs such as tie density to be reduced. Another important boundary condition in our theory is that we have assumed that most focal firms have sufficient connectedness to form dense multiparty collaborations; while this assumption widely holds in the VC context, it may be less likely to hold in contexts with a much larger number of firms and thus with loose connections and where there is a greater degree of entry and exit by potential partners. We also note that, while our status disparities arguments highlight general benefits to such status differences, the positive benefits of facilitating different equity ownership may be especially pronounced in our focal context, and thus further research is needed to examine the extent to which such status disparity dynamics generalize to other multiparty contexts.

FUTURE RESEARCH DIRECTIONS

While offering many insights into VC syndication and multiparty collaborations, our paper does have some limitations and a bounded scope that future research may be able to address. First, while we have built on our qualitative fieldwork to assume that syndicate formation requires mutual agreement amongst all parties, we have not considered how

differences in the assumed responsibilities of each VC after the investment may further influence syndicate formation. While likely requiring further adaptation of our empirical methods to include assumptions about each VC's responsibilities, this seems an interesting direction for future research—for instance, building on recent work by Garg (2013) to explore the impact of which VCs might get board seats on syndicate formation. Second, due to both empirical limitations and for theoretical simplicity, we have not considered how the quality of prior ties influences the effects or efficacy of syndicate density, and this is another promising direction for future research. Third, out of a desire to highlight the importance of group-level constructs on syndicate formation, we have not considered the role of entrepreneurs in constructing syndicates with these different attributes. Here, our logics may be extended to consider how some structures may actually give entrepreneurs greater influence (i.e., sparser syndicates), and we believe this is another interesting avenue for future research. Overall, our paper highlights the importance of a group-level perspective on multiparty collaboration formation. It is our hope that future research may build on this theory and these methods to explore how a variety of other group factors influence multiparty collaboration in both VC syndication and in other contexts.

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